Atech-crab v6.3

A four-legged walking robot: Dorian Todd's open quadruped Sesame, refitted with a new deck and lid for the Atech 14-port board and screen, driven by eight MG90S servos.

New in v6.1 to v6.3

Snap-fit deck	The plate clicks onto the spine and the head with six snap hooks. Enough to try it; no tools.
Screws for strength	4 screws from above into columns on the spine and the head. Every screw gets 4 mm of thread.
One screw for the body	M2 × 6 pan-head self-tapping everywhere. Only the servo horns take M2.5 × 5 machine screws.
Wire routing	A lead slot per servo through the internal frame, lead holes in the spine, clip-on wire bobbins on the lid.
Frame screw holes	Wider, so stock M2 heads go down into them.

Print

PLA, 0.2 mm layers, 0.4 mm nozzle, 2 walls, 15 % gyroid, tree supports from the build plate only. The STLs are already the right way up. On an H2S, H2D or A2L all 30 parts fit on one plate (about 6 h, 162 g). Sliced Bambu files for 13 printers are in the repo.

Status: designed and checked in CAD (Atech Atelier's fit check). Not yet printed or test-fitted: print the deck first and check that the hooks click.

Screws to buy

Screw	For	Need
M2 × 6 mm pan head, self-tapping, Phillips, head Ø4.5	the whole body	~55
M2.5 × 5 mm machine screw	servo horns	8

The latest in the repo

```
github.com/atechdotdev/ara-crab
branch main, commit bd4e0c3
```

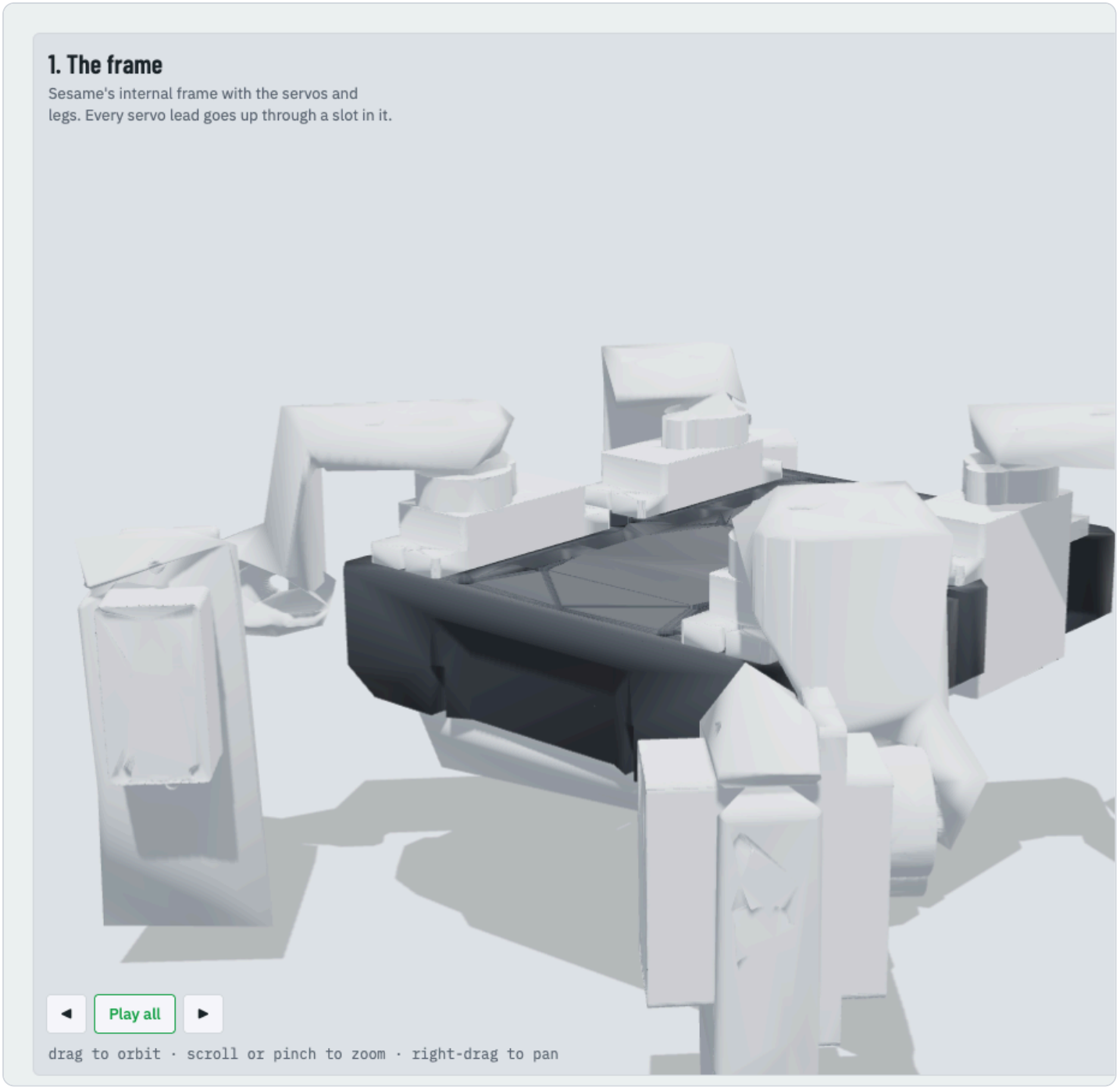
Assembly tutorial: `atech/README.md` (Assemble)
Print guide: `atech/print/README.md`
STLs: `atech/print/stl/`, wire parts `atech/print/stl_wires/`
Bambu files: `atech/print/gcode/`
CAD (Atech Atelier / FreeCAD): `atech/cad/sesame_atech_wired.py`

Licence

Apache-2.0. A derivative of the Sesame Robot Project by Dorian Todd (github.com/dorianborian/sesame-robot); see LICENSE and NOTICE.

The animation

Open **Atech-crab-assembly.html** in a browser (online, for the 3D library) to watch every step in 3D, with the screws driven in and the hooks clicking. Turn the screws on or off to compare.



Step 1

The frame

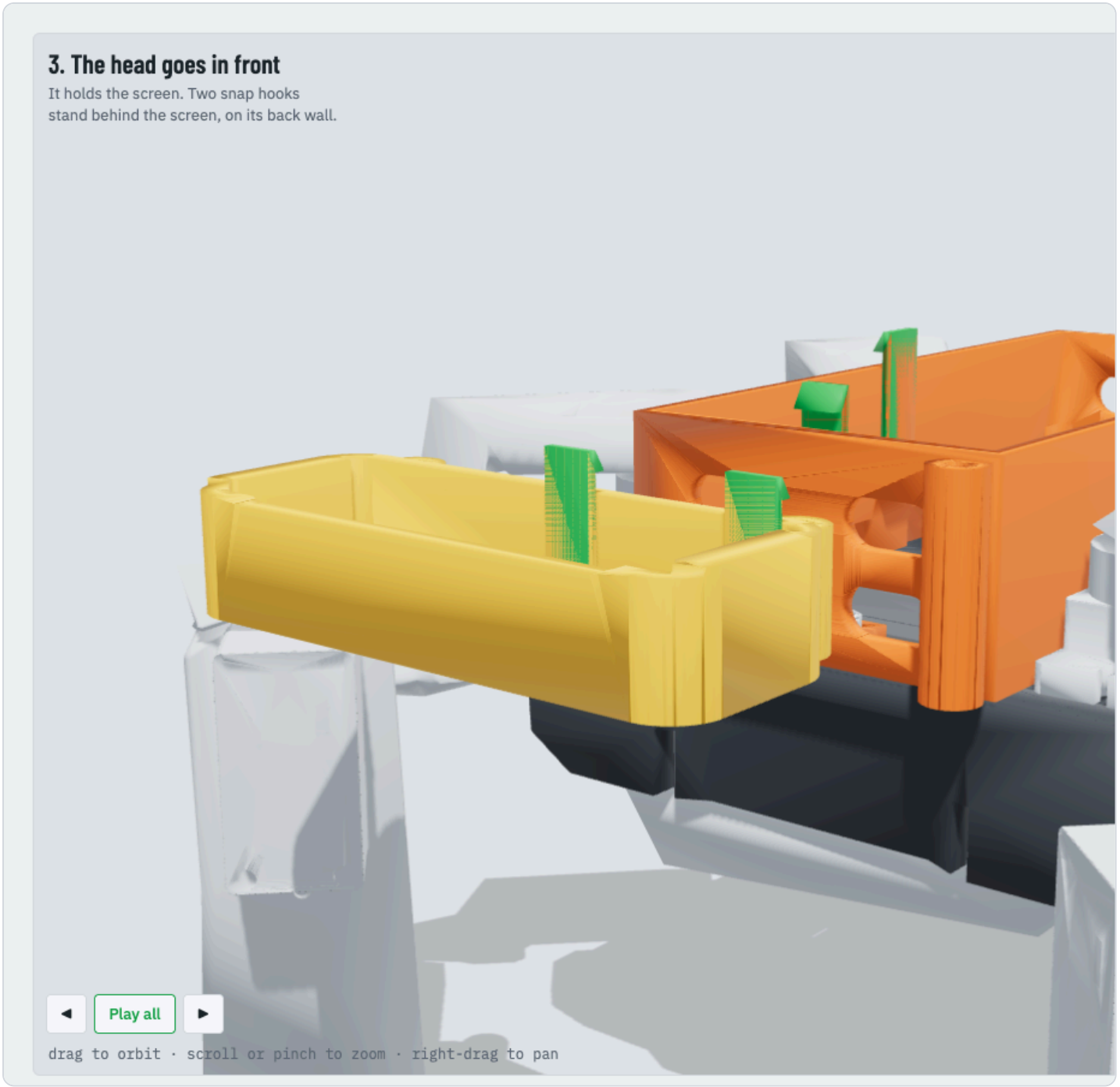
Sesame's internal frame with the servos and legs. Every servo lead goes up through its own slot in the frame, from the wire channels underneath into the spine.



Step 2

The spine sits on the frame

Screwed from below through the frame's 4 corner holes (head pocket 6.0 mm, so stock heads go in) and from above through 2 pads at the spine's ends.



Step 3

The head goes in front

It holds the screen. Two snap hooks stand on its back wall, behind the screen, where they miss both screen ribbons.



Step 4a

The plate goes on

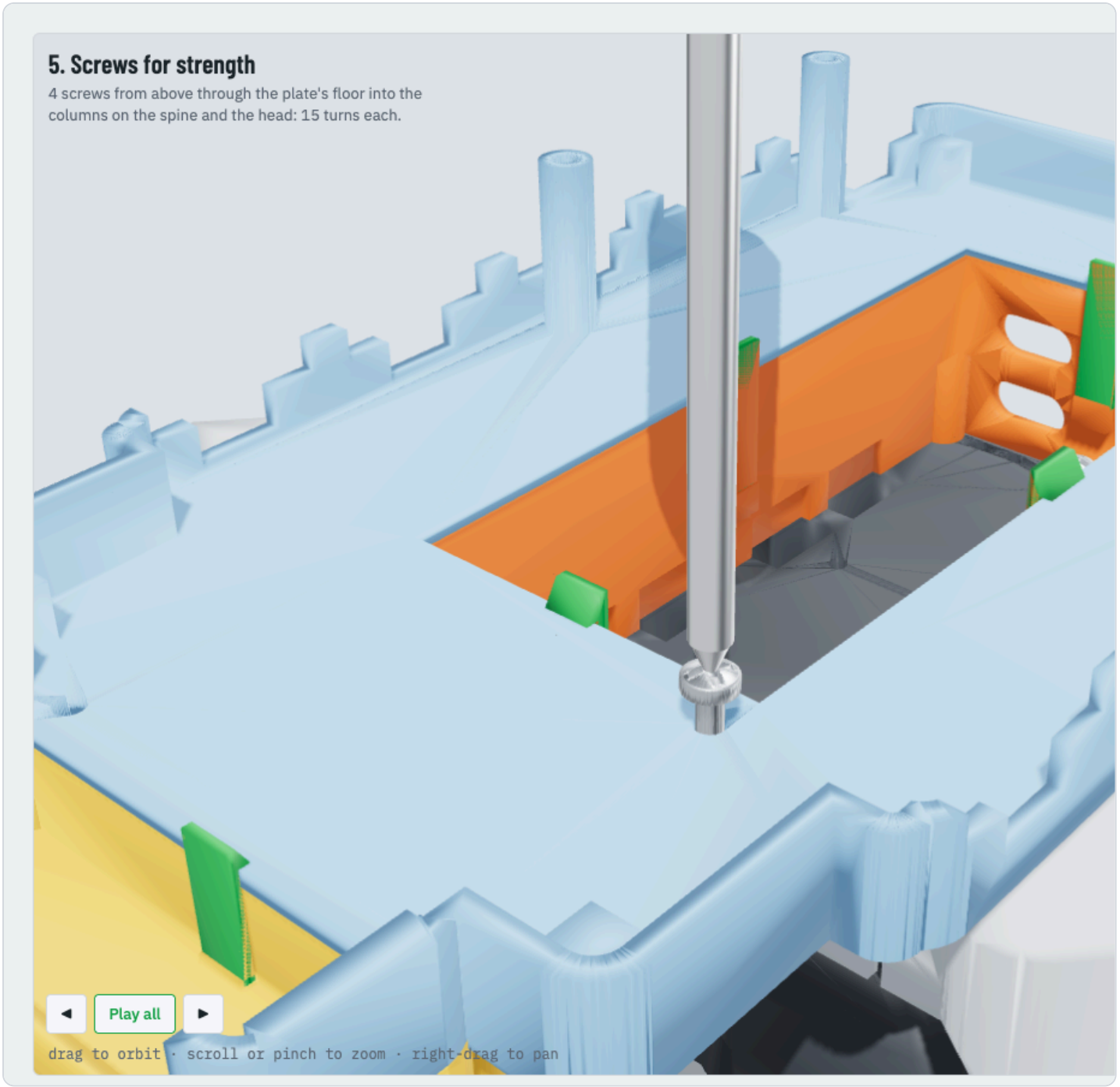
Pressed down, the edge of the plate's floor slides down each hook's ramp and bends it inward, about 6° (1.0 % strain on the spine's 12 mm hooks, 1.3 % on the head's).



Step 4b

Click: it holds

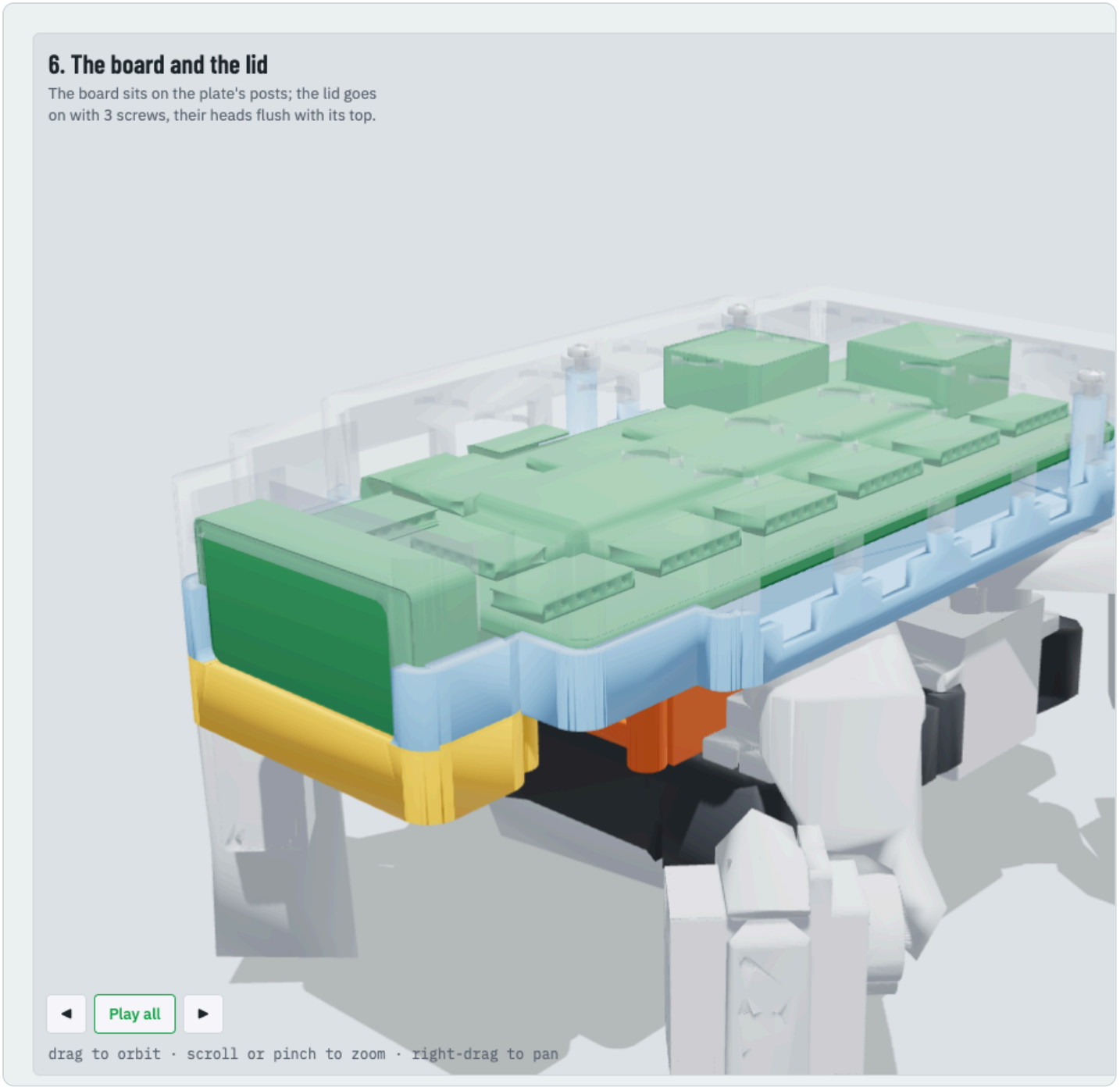
Past the ramp, all six hooks snap back over the floor's edge. The 0.55 mm keys stop the plate sliding; the hooks stop it lifting. Enough to try it, no tools.



Step 5a

Screws for strength

4 M2 × 6 screws from above through the plate's floor, into columns that stand on the spine and the head (they print upright, no supports). 15 turns each, 4 mm of thread.



Step 6

The board and the lid

The board sits on the plate's posts. The lid goes on with 3 of the same screws, their heads flush in pockets, so the wire bobbins still clip on over them.